EXHAUSTIVE OFFICIAL USER GUIDE • VERSION V1.4.16 (2026)

RandoTracker

Collaborative Live GPS Tracking Platform for Hiking Clubs, Multi-Level Groups & Outdoor Mountain Safety.

AUTHOR & DESIGN
Jean-Luc DAUSSY
Official 2026 Edition

APPLICATION TYPE	MULTI-TRACK GPX	TOPO CARTOGRAPHY	RESCUE & SAFETY
Web PWA & Android (Play Store / APK)	**1 to 5 synchronized parallel circuits**	**IGN France, Esri Satellite, Swisstopo, UK Topo, Slopes >30°**	**SAMU 15, 112, 999, 114 SMS & GPS DD / DMS**

🎄 1. Overview & Multi-Level Group Management

In hiking clubs, mountaineering associations and ramblers groups, outdoor outings routinely gather dozens of participants divided into **multiple pace groups**: Discovery / Easy loop (7 to 10 km), Intermediate loop (12 to 15 km), and Expert / Sport trek (18 to 25 km). RandoTracker is the only purpose-built platform that unites all groups on a single high-security map.

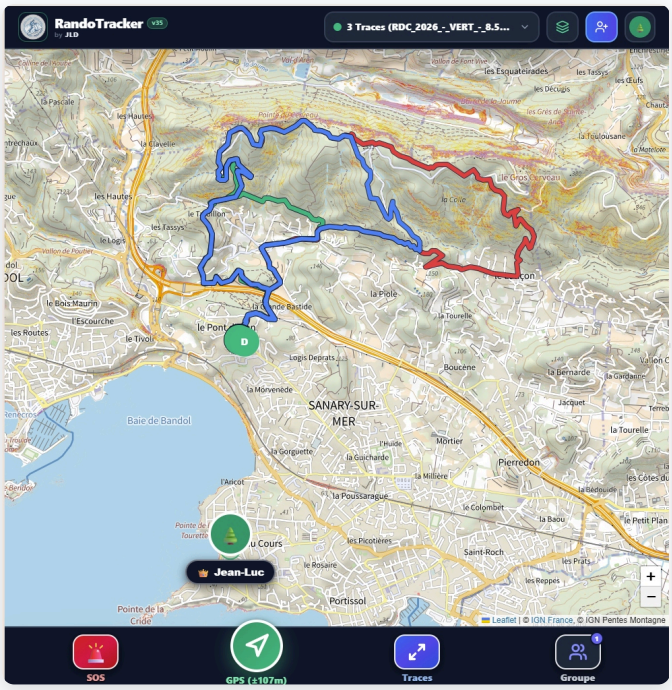


Figure 1: Official Panoramic View — Gros Cerveau Massif & Sanary-sur-Mer: 3 simultaneous GPX circuits (Green 8.5 km, Blue 11.3 km, Red 17.0 km), **Steep Slopes Overlay (>30°)** in orange/yellow, departure point "D", Leader 👑 **Jean-Luc** and **4 Giant Bottom Keys** (🚨 SOS, 🎯 GPS, 🗺️ Tracks, 👥 Group).

🎯 Simultaneous Multi-Group Tracking & Parallel Tracks

Load up to **5 GPX tracks in parallel**. Each team follows its route with a dedicated color. When multiple circuits share the same trail segment, RandoTracker applies the **PolylineOffset** algorithm to automatically display parallel offset tracks without overlaps.

⏱️ Automatic ETA Calculation & Live Telemetry

Automatic binding of each walker to their chosen track. Real-time computation of progress, remaining distance, current speed and **accurate Estimated Time of Arrival (ETA)** at the trail end, making group carpool rendezvous effortless.

🔒 Absolute Privacy & Zero Cloud Storage

No account required, no passwords, zero advertising tracking. Operates on an ephemeral "Walkie-Talkie" model: GPS coordinates flow in real-time RAM and are never permanently recorded on any central database or on other members' smartphones.

📱 Outdoor Ergonomics & Enhanced Readability

High-visibility outdoor interface: 4 giant bottom buttons usable with wet fingers or winter gloves, text scale selector (+15%, +30%), and draggable dialog boxes (*Drag & Drop*) to prevent covering key trail junctions.

💡 **Golden Rule for Club Leadership:**
The lead guide takes the role 👑 **Hike Leader / Guide** at the front. The co-leader closing the group selects 🛡️ **Sweeper / Tailguide**. Regular participants choose 🥾 **Hiker**. This allows everyone to spot the head and tail of the group at a single glance.

🗺️ 2. Topo Cartography, Esri Satellite, Aligned Map Tools & Elevation Profile

RandoTracker bundles world-class topographic maps, advanced elevation analysis and a precision floating map toolbar:

🏔️ Official Topo Maps & Esri World Imagery

- **IGN France Plan v2 Topo**: Marked GR/PR trails, contour lines, landmarks, and shelters with sharp Retina rendering.
- **Esri World Imagery (HD Satellite)**: High-resolution global aerial imagery, ideal for inspecting tree cover and ridge terrain.
- **Swisstopo & UK Topo**: Official Swiss and Great Britain maps with auto-detection based on current coordinates.
- **IGN España MTN Topo**: Perfect for Pyrenees and cross-border expeditions.
- **Steep Slopes Layer (>30°)**: Vital orange/yellow security overlay for avalanche prevention and rocky dropoffs.
- **Dynamic Metric Scale Bar**: Continuous scale display (50m, 100m, 500m, 1km) at the bottom left.

🧭 Floating Map Toolbar Aligned with Zoom (+ -)

- 🧭 **Google Maps Style Compass with Rotating North**: 3D faceted crimson needle with bold white **"N"**. Rotates dynamically with phone magnetometer and GPS heading. Tap once to instantly snap map to North.
- 📍 **Geocoded Waypoint Marker**: 1-tap button to drop a persistent point of interest at your GPS location (Parking, Water Spring, Hazard, Viewpoint, Picnic, etc.). Saves locally and syncs over MQTT in real-time to all group devices!
- 🎯 **Instant GPS Recenter**: Centers and fits the map to your exact position with one click.
- **Strict Geometric Alignment**: All 3 buttons match Leaflet's zoom (+ / -) width (34px) and vertical center axis.

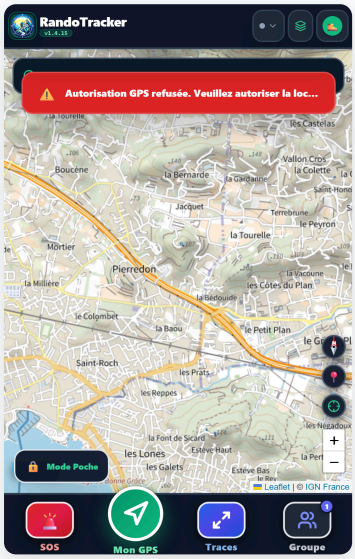


Figure 2: Live Map — Aligned floating tools (🧭 Compass N, 📍 Waypoints, 🎯 Recenter), waypoint pins and metric scale bar.



Figure 3: Cartographic Layers — IGN France, Esri Satellite, Swisstopo, UK Topo, IGN Spain and Slopes >30°.



Figure 4: Mark a Waypoint — Icon picker (🅿️ Parking, 💧 Water, ⚠️ Hazard...), custom label and live group broadcast.

📈 Zoomable Elevation Profile (1x to 12x), Live Scrubber & Parallel Tracks

- **Progressive Zoom 1x to 12x**: Interactive magnifying lens to analyze steep ascents, mountain passes, and flat breathers.
- **Live Synchronized Scrubber**: Sliding your finger across the elevation profile lights up the exact location on the map.
- **Cumulative Gain (+D / -D) & Grade**: Accurate calculation of total climb, descent, min/max altitude and slope gradient.
- **Parallel Track Offsetting**: Prevents overlapping trails when several teams share the same path.

📶 Offline Map Pre-Caching at Home (100% Offline)

The Wi-Fi Habit before leaving: Open RandoTracker, import your GPX and pan across the trail zooming in (levels 14 to 16) for 30 seconds. Map tiles are permanently saved in local cache (`rando-tiles-v1`). Even in deep zero-reception canyons, the map renders at 0 ms!

🔍 Global City & Summit Search with Country Flags

Integrated search bar with international autocomplete: type *Ben Nevis* GB, *Chamonix* FR, *Zermatt* CH or *Ordesa* ES to instantly pan and auto-switch to that country's official topographic map.

👥 3. Trailhead Onboarding, My Profile, Stop Tracking & Group Telemetry

RandoTracker revolutionizes group gathering at the trailhead: zero account setup, zero passwords, instant 3-second QR pairing!



Figure 5: Trailhead Invite — Ephemeral QR Code with automatic synchronization of tracks and waypoints.

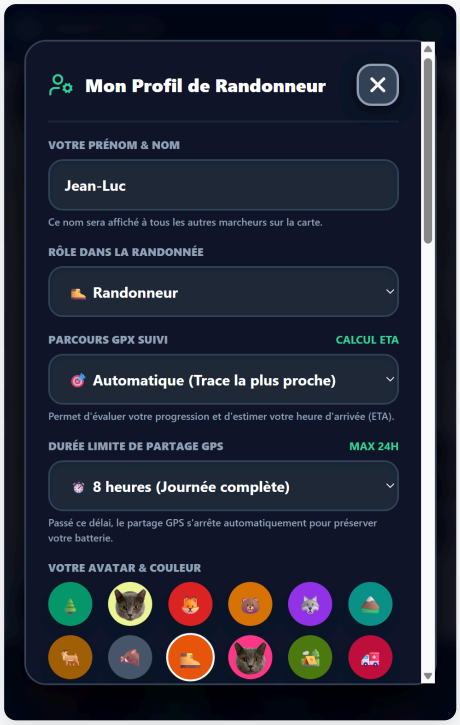


Figure 6: My Profile — Avatar picker, roles, sharing timer and red button 🔴 **Stop Tracking & Close App**.

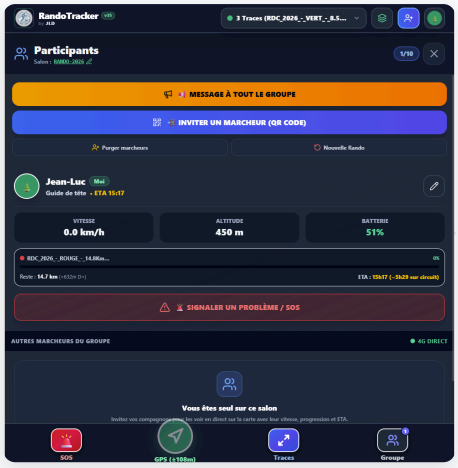


Figure 7: Participants Drawer — Live speed, altitude, battery %, cross-track deviation and dynamic ETA.

📲 1. Instant Trailhead Onboarding & Auto-Sync

- The leader opens **"Invite"** showing the secure session QR Code.
- Each walker scans the code using their phone camera (or the app's built-in QR scanner): joins the room in 1 click.
- **Automatic Track & Waypoint Sync**: New joiners instantly receive all active GPX circuits and waypoints without handling any files!

🥾 2. User Profile & Complete Stop Tracking Control

- **First Name & 12 Outdoor Avatars**: Hiker, Yellow Cat, Pink Cat, Wolf, Fox, Bear, etc.
- **Role in the Group**: Hike Leader 👑, Co-leader 🥈, Sweeper 🛡️, Regular Hiker 🥾, First Aid 🩺.
- **GPS Sharing Duration (2h, 4h, 8h default, 12h, 24h)**: Automatic countdown to preserve battery and privacy.
- 🔴 **Stop Tracking & Close Button**: Immediately terminates GPS watching, releases screen wakelocks, and shuts down all background processes.

👥 3. Participants Drawer & Inter-Hiker Distance Matrix

- **Live Telemetry Dashboard**: Traveled distance (km), current speed (km/h), elevation (m) and battery level (🔋 %).
- **1-Tap Inter-Walker Distances**: Tap any hiker on the map to instantly reveal exact straight-line distances to every other teammate.
- **Cross-Track Warning & Audio Chime (>50m)**: Instant audio/visual alert if a participant wanders off the designated trail.
- **Dynamic ETA calculation**: Personalized arrival time calculated against terrain slope and actual walking pace.

📍 Waypoint Persistence & Live MQTT Synchronization

All dropped waypoints (parking, drinking water, hazards, viewpoints) persist across app closures and browser reboots in local storage. When switching rooms or starting a new hike, they are automatically shared with everyone in the room via real-time MQTT.

🛡️ Sweeper Safety Role on the Trail

The sweeper monitors cross-track distances and battery levels in the Participants Drawer: if a hiker strays into yellow or red alert, or battery drops below 20%, the sweeper can contact them or broadcast an immediate audio callout.

🚨 4. International 50/50 Emergency SOS Dispositif & Broadcast Messaging

In case of injury or distress in mountain terrain, every second counts. RandoTracker incorporates a 50/50 split emergency console accessible in 1 tap on the red 🚨 **SOS** button:

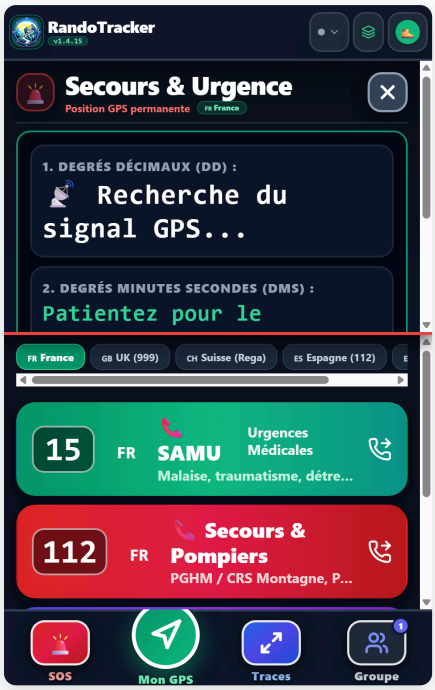


Figure 8: SOS 50/50 France — Pinned GPS coordinates (DD/DMS), phonetic voice dictation, **15 SAMU**, **112** and **114 SMS**.



Figure 9: Spain Tab ES — Automatic emergency numbers switch: **112** and **061 Urgencias**.

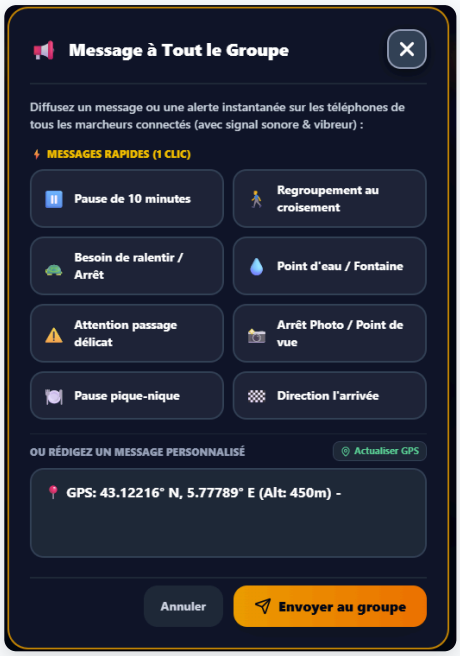


Figure 10: Group Broadcast — 8 quick 1-tap presets with audio chime, vibration alert and embedded GPS link.

🚨 High-Security 50/50 Multi-Country Rescue Console:

- **Fixed Top Half (Immediate Location)**: Giant GPS coordinates in both **Decimal Degrees (DD)** and **Degrees Minutes Seconds (DMS)**.
- **Phonetic Voice Dictation Guide**: Pre-formatted script ready to be read aloud to emergency dispatchers without inversion mistakes.
- 📋 **Copy GPS Button**: Instant copy to clipboard for SMS or satellite messengers.
- **Dedicated Country Tabs**: GB UK (999, 112), FR France (15 SAMU, 112, 114 SMS), CH Switzerland (144 Medical, 1414 REGA Mountain Rescue), ES Spain (112, 061), EU Europe (112).
- **1-Tap Direct Dialing**: Calls emergency services immediately without leaving coordinates screen.
- 🚨 **Group SOS Siren**: Triggers alarm and strong vibration on all club members' phones.

📢 5. Group Broadcast Messaging & Fast On-Trail Coordination

📢 1-Click Broadcast Messages for All Walkers

- **8 Key Outdoor Presets**: "⏸️ *10-minute break!*", "🚶 *Regroup at summit!*", "⚠️ *Caution, slippery rocks*", "🐢 *Need to slow down*", "💧 *Water spring refuel*", "🥪 *Lunch break*", "🔄 *Route change*", "👍 *All accounted for*".
- **Audio Chime & Vibration**: Wakes up phones inside backpacks or jackets.
- **"View on map" Button**: One tap centers the map directly on the sender's location.

🌲 Total Zero-Reception Resilience & Local Caching

- **100% Autonomous GPS Chip**: Satellite positioning operates completely without cellular coverage.
- **5-Hour Buffer Memory**: Last known positions remain visible with an indicator showing data age.
- **Silent Auto-Reconnection**: As soon as any phone catches signal on a ridge, all pending positions and waypoints sync automatically.

⚠️ Universal Trailside Accident Protocol:

1. **Protect:** Secure the scene and casualty (survival blanket, shelter from cold and wind).
2. **Alert:** Open 🚨 **SOS** on RandoTracker, read DMS coordinates to **999** or **112**, state injury details.
3. **Assist:** Deliver first aid while awaiting mountain rescue or helicopter dispatch.

🔋 6. Record Battery Autonomy, Background Service & Pocket Mode

Battery endurance is the #1 safety asset in the wilderness. RandoTracker is engineered for ultra-low power consumption:

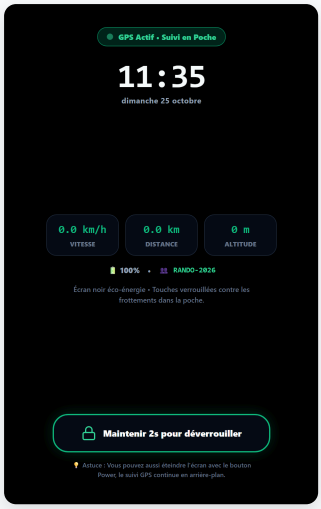


Figure 11: Pocket Mode (🔒) — Pure OLED black theme, giant clock, anti-touch protection and 1.5s long-press unlock.

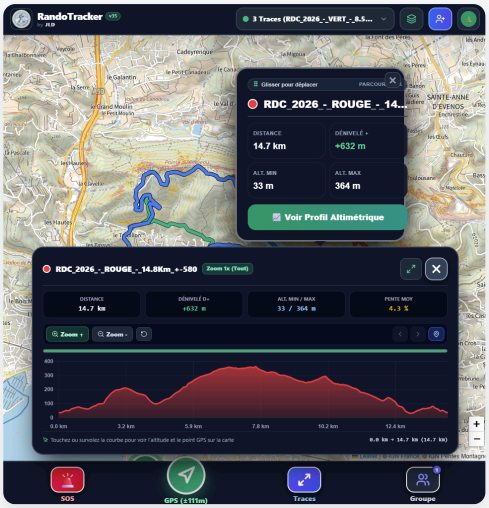


Figure 12: Live Elevation Profile — Zero-drain vector graphic engine preserving phone power throughout the day.

🔋 Real Field Test: < 1% Battery Consumed in 5 Hours!

Certified field measurement during a full-length mountain outing:

• **1h22** active screen map consultation + **3h38** pocket background tracking = **less than 1% battery consumed**!

• Full day of hiking (8 to 10 hours): only **2% to 3% total battery drain**.

🔒 Android Screen-Off Service & Web / iOS Pocket Mode

• **Native Android App:** Simply turn off screen with physical power button. Certified Google Play background service maintains GPS with zero complex setup.

• **Web & iPhone (PWA):** Tap **Pocket Mode (🔒)**. Displays OLED black screen with giant clock and 1.5s hold-to-unlock safety.

🏆 7. Exhaustive Feature Comparison: RandoTracker vs Alternatives

Features & Key Criteria	RandoTracker (2026)	AllTrails / Komoot	WhatsApp / Live Location
Installation & Access	✅ **Instant Web PWA or Android App**	⚠️ Mandatory App Store installation	⚠️ Third-party app required
User Account & Privacy	❌ **Zero account / 100% ephemeral**	⚠️ Email + Password + Ads tracking	⚠️ Phone number visible to all
Offline Map Pre-Caching	✅ **Free home Wi-Fi caching (0 ms)**	❌ Paid feature ($30 to $40 / year)	❌ No map caching capability
Battery Efficiency	✅ **Record: <1% over 5h • OLED eco**	⚠️ Heavy drain (ads, white themes, SDKs)	⚠️ High battery drain in live share
Multi-Track GPX in Parallel	✅ **1 to 5 GPX tracks simultaneously**	❌ 1 active track only	❌ No GPX support
Parallel Tracks (PolylineOffset)	✅ **Automatic offset on shared paths**	❌ Tracks overlap and hide each other	❌ Not available
Google Maps Compass & Waypoints	✅ **3D N needle + MQTT synced pins**	⚠️ Basic compass / No live pin sync	❌ Not available
Official Topo Maps & Esri Satellite	✅ **Included & 100% Free (IGN, Esri, UK, CH)**	❌ Expensive annual subscription	❌ Crude standard road maps
Slopes >30° & Dynamic Metric Scale	✅ **Included (Steep slope safety + Scale)**	❌ Locked behind Premium tier	❌ Not available
Zoomable Elevation Profile	✅ **1x to 12x Zoom + Gain + Scrubber**	⚠️ Static non-zoomable profile	❌ Not available
Group Telemetry & Pack Tracking	✅ **Entire pack, Roles, Avatars & ETA**	❌ Individual tracking only	⚠️ Contact list without hiking telemetry
50/50 Multi-Country SOS Console	✅ **GPS DD/DMS + 999 + 112 + 15 + 114**	❌ No dedicated emergency module	❌ No formatted GPS coordinates
Full Stop Tracking & Exit Button	✅ **Dedicated 🔴 Stop & Exit button**	⚠️ Persistent background service	⚠️ Must remember to stop sharing manually

📋 8. Step-by-Step Club Outing Checklist

1️⃣ Before Leaving (At Home on Wi-Fi)

1. Open RandoTracker and load GPX files (**"TRACKS"** $ ightarrow$ **"+ Import"**).
2. **Map Pre-caching**: Pan and zoom (levels 14-16) along route for 30s to store Topo HD tiles.
3. Assign distinct colors to each circuit (10-color outdoor palette).

2️⃣ At the Trailhead Parking (Leader & Group)

1. Leader opens **"Invite"** and presents the room's **QR Code**.
2. Walkers scan the code: joins room and automatically receives tracks and waypoints.
3. In **"My Profile"**, enter first name, select role, assign track and tap **"🎯 MY GPS"**.

3️⃣ On the Trail (Navigation & Safety)

1. Monitor group pace gaps and ETAs in the Participants Drawer.
2. Use **"Broadcast Message"** to announce regroupings or stops with automatic GPS.
3. Mark key landmarks, springs, or hazards using the 📍 **Waypoint** button.

4️⃣ Returning to Trailhead (End of Hike)

1. Verify all participants have safely reached the parking area.
2. Open **My Profile** and tap the red 🔴 **Stop Tracking & Close App** button.
3. Session clears automatically: zero residual data stored anywhere.

🛠️ 9. Comprehensive Troubleshooting Guide (What to do if...)

📍 1. GPS stops updating when phone is in pocket

- **Android Battery Setting:** Go to *Settings $ ightarrow$ Apps $ ightarrow$ RandoTracker $ ightarrow$ Battery* and select **"Unrestricted"**. Avoid "Optimized" which suspends standby GPS.
- **Precise Location:** Under *Settings $ ightarrow$ Location $ ightarrow$ RandoTracker*, ensure **"Use precise location"** is enabled.

👥 2. Cannot see other walkers in the room

- **Same Room Code:** Ensure all teammates have entered the exact same room code (e.g., `RANDO-2026`).
- **Refresh Button:** In the Participants Drawer, tap the 🔄 icon to force immediate MQTT resynchronization.

📢 3. No sound chime for group broadcast messages

- **Audio Unlock:** Tap the screen at least once upon launching the app to authorize browser Web Audio synthesis.
- **Do Not Disturb:** Check that your phone is not in total silent mode or "Do Not Disturb".

🗺️ 4. Blank or grey map tiles in remote mountains

- **Wi-Fi Habit:** Always remember to pan and zoom over the route at home before heading out.
- **Layer Switch:** Open the layers menu ( ) and switch to **Esri Satellite** or **OpenStreetMap** which frequently hold cached tiles.

🔒 10. Google Play Store Certified Security & Zero-Cloud Privacy

🛡️ 1. Zero Malware & Play Protect Certification

100% Safe App Delivery: Every release (.aab) undergoes automated deep antiviral scans by Google Play Protect: zero malware, zero spyware, zero tracking bloatware. Cryptographically signed.

☁️ 2. Zero Central Database & Full GDPR Compliance

Ephemeral Architecture without History: No GPS coordinates are stored on a cloud server. Positions flow live through volatile memory (RAM) and disappear instantly. Settings remain strictly on your own device.

RandoTracker — Group Hiking Tracker, Multi-Pace & Topo Cartography
Progressive Web App (PWA) & Android App engineered for outdoor club safety, leadership and conviviality.

CONCEPT & DEVELOPMENT
Jean-Luc DAUSSY — 2026